

A Review: Information Extraction Techniques From Research Papers

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Abstract— Text extraction is a crucial stage of analyzing Journal papers. Journal papers generally are in PDF format which is semi structured data. Journal papers are presented into different sections like Introduction, Methodology, Experimental setup, Result and analysis etc. so that it is easy to access information from any section as per the reader's interest. The main importance on section extraction is to find a representative subset of the data, which contains the information of the entire set. Various approaches to extract sections from research papers include stastical methods, NLP, Machine Learning etc. In this paper we present review of various extraction techniques from a PDF document.

Keywords— *Information extraction; NLP; Machine Learning; PDF*

I. INTRODUCTION

Data is the basic means of representation of facts in an understandable manner which can be globally accepted. Data is required for the flow of information such that, it provides a platform for expressing the views and specific perspectives of an individual. It is the essential unit of communication over the generations which has facilitated in the need for preserving and utilizing the data. Over the years, there has been a rapid escalation in the information gradient supported by convenient methods of data storage and representation based on the underlying application. From this abundant data the main challenge encountered by a user is to capture data of his interest. Retrieving a part of the data or extracting a relevant section from a given document is an active area of research and lead to the introduction of search techniques based on Key terms and pattern matching.

The research document which are usually stored as textual documents are in a Portable Document Format (PDF) consists of huge amount of significant data, which is the primary source of information reserves for the research scholars. These documents provide a base on which existing information is extracted, analyzed, retrieved in case of manipulations if necessary, and stored as well for further reference. In order to increase the readability of the document, it is being categorized into different sections where a very brief insight into the document is provided in terms of abstract and

introduction. This is further followed by several other sections like literature review, methodology applied, results and conclusion.

Key terms are the exact data items which are explicitly defined in terms of other data units. The use of key terms has enabled the retrieval of required data in much better manner. More complex applications resulted in more number of key terms, finally resulting in the usage of key terms with associated index terms called as values. The key - value pairs increased the effectiveness of data retrieval in a proficient manner. The selection of key terms plays a very significant role as the selection process is based on the data semantics and syntax which varies over textual, audio and video data representation. This enables in highlighting the characteristic features of the data document and providing the fundamentals towards Information Extraction or Information Retrieval. A list of key terms associated with every document enable the user in streamlining the documents based on their applicability. The noble example of information extraction based on key terms is the Google Search Engine which works on the principles of PageRank algorithm using a logarithmic scale [1]. The main advantages of PageRank are less query time, less susceptibility to localized links, more efficiency. On the other hand disadvantages of PageRank are may retrieve documents less relevant to user query, rank sinks, it is a static algorithm, dangling links [2]. The real time applications of today not only need information retrieval to happen in a very efficient manner but always contemplate more on the gathering data on qualitative basis than on quantitative levels. As the data size has increased tremendously, manual information retrieval has become impossible. Hence it provides concepts to model this data so that they can return all the documents that are relevant to the query term and ranked based on certain importance measures.

A much broader categorization of the documents can be given in terms of free-text, semi-structured text and structured text [3][4]. A free-text is a loose collection of scripts and stories which are not well formatted and have very minimal connectivity established. Extracting information from such documents is a very tedious job as each unit of data that is present in the document must be analyzed to understand its

existence. The data that is maintained as a structured text is stored in an organized manner and retrieved by their key values. The semi-structured text on the other hand follows a domain level categorization based on automated method as proposed by N. Ashish et.al [5]. This paper provides the discussion of methods for information extraction in general and then focus on the review of section extraction methods in particular research documents. The proposed survey provides a detailed insight into the existing information extraction, and their shortcomings. Section II gives the general Information Extraction methods followed by specific extraction for PDF documents in Section III. Section IV provides the category analysis of extraction methods and Section V gives the conclusion.

II. GENERAL EXTRACTION METHODS

Information Extraction (IE) is the process of extracting useful data from the already existing data by employing the statistical techniques of Natural Language Processing (NLP) [6]. It is defined as the act of identifying, collecting and regularizing relevant information from the given text and producing the same in a suitable output structure [7]. Although the extraction process has been automated over years, the need for training the system to work as per the rapid changes within the specified time range is very much important.

The process of extraction can be decomposed to make its functionality simpler by following a modular approach for each of its task of selection, reordering and composition. Each of the sub-process thus identified can in turn adopt a flexible technique based on the underlying application and this is advantageous in case of modular approach. It also facilitates in improving the validity of the system, relationship and level of coupling between the components. The major modules identified under IE are: segmentation, classification, association, normalization and co-occurrence resolution. For the text given as the input, segmentation focuses on dividing, based on the semantics and syntax of the data format. Lexical analyzers are commonly used to define the semantic rules to enable an effective data segmentation process. The basic data units are identified by segmentation and of all the identified data units, specific data is given primary importance based on the priority, number of times it is being repeated and their usage according to the semantics of the language defined. The most commonly used segmentation technique is by the Viterbi Algorithm [8] which works on the concept of state machines. Machine learning approaches [9] are employed based on the probabilistic rules followed by the Context Free Grammars. Classification methods follow segmentation which categorize similar data units into a common group such that meaningful relationships can be constructed between the grouped entities. The normalization module is used to confirm that the extractions are according to the specified formats to enable synchronization between the forms of data obtained at each module. In case of repetition, co-occurrence [10] is used such that identity of each data entity is maintained. Classification in extraction enables us to classify the major domains into which the segmented data is finally stored by using machine learning approaches. Common scenario used for representing the

classified data is by the use of decision trees. In order to extract the related entities, association rules are used to extract the desired relations of various categories.

Machine learning plays a very critical role in IE as we are able to achieve most accurate results with very less error propagation rate. In case if any improvements are required then Machine Learning with NLP provides a basis for the proofs that are obtained through empirical methods. Extraction [11] through Machine Learning if automated helps in the extraction of patterns covering different areas in very minimal time duration. Commonly used Machine Learning based Extraction methods can be categorized as supervised learning and unsupervised learning. Supervised learning is used when there is user interaction in defining the rules for the extraction process, or in defining the sample training example based on which the supervision task can be carried out. The categories of supervised learning are propositional learning and relational learning. Using the core mathematical foundations, propositional learning is practiced where the examples are represented in terms of zero order logic or attribute-value logic. On the other hand, relational learning uses the first order logic to represent the examples for the learning process and is mostly useful in case of textual data type.

The need for human interaction to provide the training example is one of the major drawback of the supervised learning technique which has paved way for the unsupervised learning mechanisms which are developed by employing corpus bootstrapping method [12] to obtain the seed rules from annotated systems. Yet another category is the semi-supervised learning technique based on mutual bootstrapping and finally Hybrid NER [13] (Name Entity Recognition) system which works by combining the randomized conditional variable and a supporting base variable. However these techniques are just not enough to solve the current tribulations faced during extraction as data identification, classification and management as the core activities which have to be given due importance [14].

III. EXTRACTION FROM SPECIFIED DOCUMENT

This section highlights how extraction process for scientific literature is different from general information extraction from PDF document. Paul Buitelaar et.al [15] in their work on Topic Extraction from scientific literature have suggested automated system that is dynamic in nature to support critical systems by extracting data from scientific applications. A pattern based approach has been employed to organize the data according to the research requirements. The extraction process depends on the pertinent skills and the semantic relations between the existing phonetics and finally statistical approaches are employed for information retrieval along with learning scheme sets defined by machine learning aspects. To bring in the support for cross domains, domain-specific linguistic patterns are used. The representation of the extracted units is done here by the use of an association network which signifies the interconnections between the units there by highlighting the area of proficiency and capability between the researchers.

The work Analyzing the Dynamics of Research by Extracting Key Aspects of Scientific Papers, proposed by Sonal Gupta in [16] characterizes the given research document into key areas like focus, domain of application and different underlying techniques used. The data extracts are obtained using the semantic extraction patterns which uses bootstrapping as the learning technique and finally representing them in the tree like structure. The different domains identified here were termed as communities and this became a novel approach as it influenced in extracting the characteristics at the semantic levels by employing bootstrapping learning technique and finally representing the output in the form of dependency tree. This work facilitated in the dynamic extraction of rich information which is worth the search as it categorizes into respective communities making the search more effective.

The task of identifying the relevant topic of importance is very difficult even in case of providing the keywords. Streamlining the major key terms is itself a tedious job and upon that moving along the related data in large texts is quite a challenge. In case if we consider multi-document data set, an automated keyphrase extraction is more useful as it helps in obtaining a precise subset of categories generally termed as clusters. The algorithm for text extraction used in this case could provide exact clusters which were much better compared to the previously used keyphrase extraction techniques irrespective of working for various domains there by providing domain independency. The cluster-to-cluster similarity index could also be calculated using the proposed algorithm. But the proposed algorithm did not employ any ranking schemes which would have enhanced the removal of unwanted permuted combinations of phrases [17]. Yet another keyword extraction technique based on the co-occurrence statistical information has been used in by Yutaka Matsuo [10] is applicable for a single document without expressing the obtained results in terms of any measurable quantities. The keywords are decided based on their repeated occurrence and frequency which is termed as co-occurrence distribution factor. By using the proposed algorithm, key term extraction is done very accurately even if the occurrence is rare but the term is important. The advantages of this method enabled text mining without the need for corpus and its effective usage in case of domain-independent keyword extraction.

Slobodan Beliga in [18] has provided a detailed review on keyword extraction and the different approaches used for keyword extraction. The importances of both supervised and unsupervised learning methods are understood along with the features of Croatian keyword extraction. Further categories of extraction include document-oriented and collection-oriented techniques through which more systematic extraction could be achieved. In case of applications with complex networks, Selectivity-based keyword extraction has been proposed as the new unsupervised keyword extraction technique which paves way for further research applications [19]. The advantages of Selective based keyword extraction is that it does not work on

any linguistic knowledge of the text document, but the proofs are obtained by the employment of statistical methods because on which manual annotations required is minimal and beneficial in case of fast computing.

The keyphrase are the representatives of entire textual documents which are responsible for enhancing the processing and retrieving the necessary tasks. If the keyphrase can be easily identified in case of research papers, then reviewing the literature can be carried out more effectively. To facilitate this, CiteTextRank has been proposed by Sujatha Das et.al [20] for the extraction of keyphrase from research articles and representing them is a graph-based algorithm. The advantages of using CiteTextRank are that it improved the precision of rating the articles based on the keyphrase and improves the efficiency and data availability in case of highly - connected networks.

There is a new method in Bo Chen [21] nonlinear model using SVM with max-margin discriminate projection is used to fine the variable and augmented variables to extract information from data set. This model is developed for supervised data set. This model can be extended to unsupervised way to cluster the data.

When we are working with data of a variety of different domains, it is very much necessary to have minimal domain specific knowledge for the automatic extraction of the key terms. If at all new domains are to be included into the system, then the earlier systems were not flexible enough for updating due to strenuous manual tuning of the domain-dependent semantics and syntax. Moreover even though we have numerous extraction techniques, arriving at the most accurate model is purely based on the design decisions made at the early steps. The complexity also occurs in terms of the overall time period taken in the extraction process. Identifying the right keyphrase and finalizing a right subset is often time consuming though automation is followed in the recent days. Analyzing and extracting data of different semantics have resulted in performance issues as well.

A. Abbreviations and Acronyms

NLP - Natural Language Processing

PDF - Portable Document Format

IE – Information Extraction

NER - Name Entity Recognition

SVM – Support Vector Machine

IV. CATEGORATION OF EXTRACTION METHODS

Extraction methods can be categorized into different classes depending on input data, analyzing techniques, information required [22]. These data can be differentiated into structured, semi-structured and unstructured data. Analyzing techniques of data analysis can be NOSQL Databases, BigQuery, MapReduce, WibiData, Skytree and Hadoop. The required information as the output can be grouped into Descriptive analytic (analytics that help in understanding what has happened), predictive analytics (analytics that help in anticipating what will happen) and

prescriptive analytics (analytics that help to respond what next). We are analyzing and reviewing the existing techniques so that a new method can be developed to overcome the disadvantages.

V. CONCLUSION

This paper discusses a few extraction methods to extract sections from documents. NLP and probabilistic extraction techniques are purely based on the design decisions made at the early steps. The complexity also occurs in terms of the overall time period taken in the extraction process. Identifying the right keyphrase and finalizing a right subset is often time consuming though automation is followed in the recent days. Analyzing and extracting data of different semantics have resulted in performance issues as well.

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